UNDERGRADUATE MECHANICS OF SOLIDS PROJECT

Compression of a Rotating Square Auxetic Lattice

Analytical model, CAD geometry audit and ANSYS simulation

10 November 2026

Abstract

A rotating-square lattice was studied using rigid-unit kinematics, a simplified ligament bending model and a displacement-controlled simulation in ANSYS Mechanical. At 5.00 mm prescribed compression, the axial support reaction was **29.895 N**. The interior axial and transverse gauges shortened by 9.614% and 6.707%, giving a face-based effective Poisson's ratio of **−0.698**. The inward transverse motion is consistent with the auxetic mechanism. The ideal periodic model predicts −1, while the finite specimen has compliant ligaments and restrained end rows.

1 Background and objective

In an auxetic lattice, the architecture can cause the structure to narrow as it is compressed. Here, square units rotate in alternating directions through flexible corner ligaments. The project derives the ideal in-plane response, audits the supplied CAD geometry and quantifies the response of the simulated specimen along the global Y axis.

Contents

Section	Page
1 Background and objective	1
2 Specimen geometry and CAD audit	2
3 Ideal rotating square kinematics	3
4 Ligament bending model	4
5 Analytical predictions	5
6 ANSYS simulation method	6
7 ANSYS results	7
8 Discussion and conclusions	11
9 Analytical calculation check	12
References	13

2 Specimen geometry and CAD audit

The global X axis is the specimen width, Y is the loading direction and Z is the through-thickness direction. The theoretical reference design and the imported STEP have the following external dimensions.

Table 1. Nominal dimensions and audited STEP bounding box.

Dimension	Nominal design	Audited STEP span	STEP bounds
Width, X	50.00 mm	40.000 mm	−6.1440 to 33.8560 mm
Length, Y	100.00 mm	79.998 mm	−9.5150 to 70.4830 mm
Depth, Z	50.00 mm	25.000 mm	0 to 25.000 mm

Open CASCADE 7.9.3.1 imported one STEP root and one fused solid. The file declares millimetres. The solid has 630 faces, 3,552 edges and a volume of 49,663.078 mm³. The bounding box measures the whole solid. Individual hinge dimensions and the active lattice footprint require separate measurements.

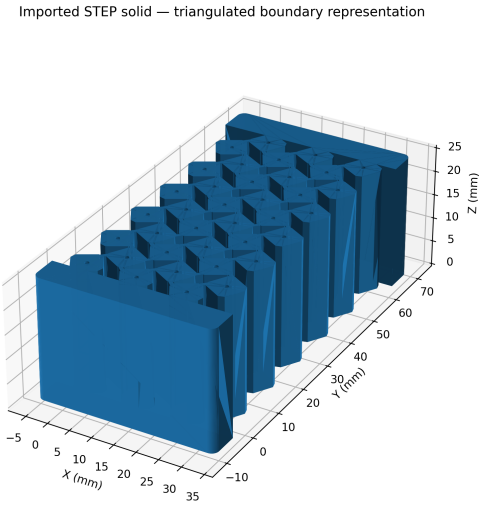


Figure 1. Isometric rendering of the audited STEP solid.

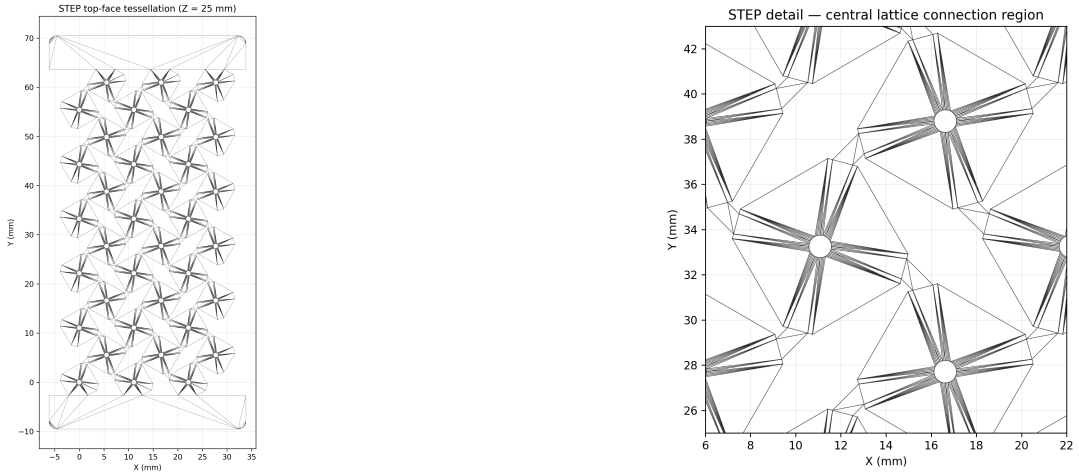


Figure 2. CAD tessellation at Z = 25 mm (left) and central connection detail (right). These figures show CAD surfaces. The finite element mesh is visible in the ANSYS exports.

3 Ideal rotating square kinematics

The nominal design has 36 square units, 18 in each orientation, at +θ and −θ. The square side a is 8 mm, the corner chamfer d is 1.2 mm and the hinge width h is 1 mm. The nominal extrusion depth t is 50 mm and the reference angle θ_o is 30°.

$$R(\theta) = [\cos\theta, -\sin\theta; \sin\theta, \cos\theta] \quad q = (a/2, a/2)$$

At a common corner, $c_A + R(\theta)q = c_B - R(-\theta)q$. Adding the two rotated corner vectors cancels the sine terms.

$$c_B - c_A = a\cos\theta\,(1, 1) = D(1, 1). \quad (1)$$

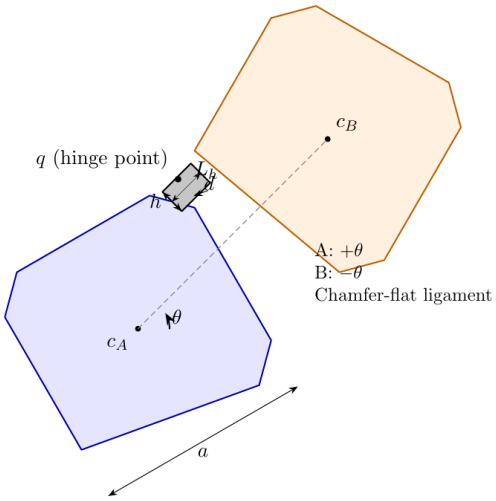


Figure 3. Corner compatibility and the finite ligament between the chamfered flats.

The second diagonal translation is $D(1, -1)$. The ideal centres are $A(i,j) = (2iD, 2jD)$ and $B(i,j) = ((2i + 1)D, (2j + 1)D)$, with i = 0–2 and j = 0–5. At θ_o, D = 6.9282 mm and the repeat pitch $L = 2D$ = 13.8564 mm.

$$\varepsilon_X = \varepsilon_Y = L/L_o - 1 = \cos\theta / \cos\theta_o - 1, \quad v_{ideal} = -1. \quad (2)$$

The engineering strain increment is $-(\sin\theta / \cos\theta_o)d\theta$, while the logarithmic strain increment is $-\tan\theta\, d\theta$. Each convention gives equal increments along X and Y. The ideal lattice scales uniformly in-plane, with zero in-plane shear [1–5].

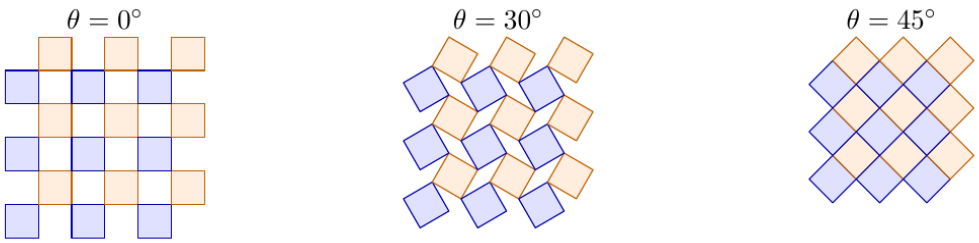


Figure 4. Ideal repeat-cell patches at 0°, 30° and 45°. The corner hinges are idealized.

Table 2. Ideal rotation and solid area fraction across the geometric range.

Axial ε (%)	-18.4	-15.0	-10.0	-5.0	0.0	5.0	10.0	15.0	15.5
Rotation (°)	45.0	42.6	38.8	34.6	30.0	24.6	17.7	5.2	0.0
Solid area (%)	100.0	92.3	82.3	73.9	66.7	60.5	55.1	50.4	50.0

4 Ligament bending model

4.1 Reference geometry

For the 3 × 6 design, the projected active extents before trimming are $5D_o + a(\cos\theta_o + \sin\theta_o)$ and $11D_o + a(\cos\theta_o + \sin\theta_o)$. Trimming the two extreme corners removes $2d \min(\sin\theta_o, \cos\theta_o) = 1.20$ mm from each extent. The nominal active width W_o is 44.37 mm and the nominal active height H_o is 85.94 mm. Two 8.53 mm end blocks overlap the lattice by 1.50 mm each, giving an overall nominal height of 100.00 mm.

4.2 Hinge stiffness and stored energy

The model treats the squares as rigid and each ligament as a straight rectangular beam of uniform curvature. It assumes linear bending stiffness and neglects shear, axial extension and local corner compliance. The nominal TPU modulus is 26 MPa.

$$L_h = \sqrt{2}\, d \cos\theta_o = 1.4697 \text{ mm}, \quad I = th^3/12 = 4.1667 \text{ mm}^4, \quad k_h = EI/L_h = 73.71 \text{ N mm/rad}. \quad (3)$$

A hinge rotates by $2\Delta\theta$ relative to its stress-free state, where $\Delta\theta = \theta - \theta_o$ in radians. The energy of one hinge is $\tfrac{1}{2}k_h(2\Delta\theta)^2$. Assigning two shared hinges per square gives $U_{sq} = 4k_h\Delta\theta^2$.

$$u = U_{sq}/(2a^2t\cos^2\theta) = 2k_h\Delta\theta^2/(a^2t\cos^2\theta). \quad (4)$$

4.3 Stress and ligament strain estimates

Differentiating Eq. (2) gives $d\theta/d\varepsilon = -\cos\theta_o/\sin\theta$. The original hinge model takes $du/d\varepsilon$ to obtain the following stress estimate based on current cell volume.

$$\sigma_h = -(\cos\theta_o/\sin\theta)\,[4k_h\Delta\theta/(a^2t\cos^2\theta)]\,(1 + \Delta\theta\tan\theta). \quad (5)$$

$$E_{eff} = Eh^3/(3L_h a^2\sin^2\theta_o) = 0.3686 \text{ MPa}, \quad \varepsilon_{lig} = |\Delta\theta|h/L_h. \quad (6)$$

The modulus follows by linearizing at θ_o. The extreme-fibre strain follows from the curvature change $2\Delta\theta/L_h$ and distance $h/2$ from the neutral axis. The nominal hinge is short, with $L_h/h \approx 1.47$, so the pure-bending assumption is a significant source of uncertainty.

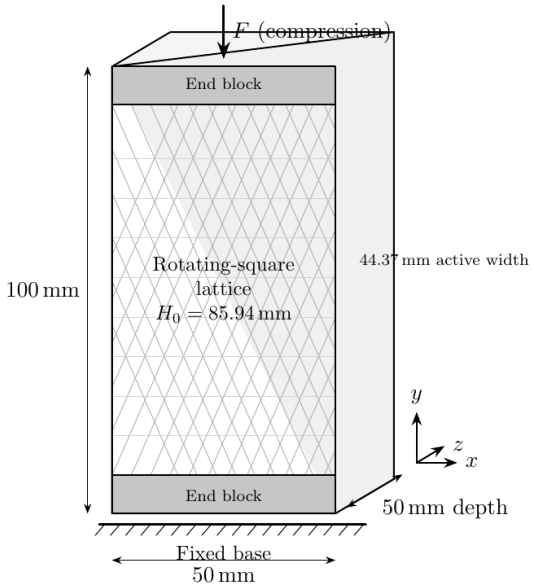


Figure 5. Nominal reference specimen and loading schematic. The dimensions apply to the analytical design.

5 Analytical predictions

A primitive cell contains one square. Its area is $2D^2$, its void area is $a^2 \cos(2\theta)$, and the solid area fraction is $\phi_s = 1/(2\cos^2\theta)$. The areal engineering strain is $(1+\varepsilon)^2 - 1$. The model opens fully at 0° and reaches geometric closure at 45°. For the nominal active height, these limits correspond to 13.29 mm extension and 15.77 mm compression.

For an imposed analytical active-region displacement ΔH, $\varepsilon = \Delta H/H_o$ and $\theta = \arccos[(1+\varepsilon)\cos\theta_o]$. The force estimate uses $|\sigma_h| W_o t$, with an active reference area of 2,218.5 mm².

Table 3. Predictions for the nominal analytical reference design.

Analytical state	Axial ε	θ	Solid area	σ_h (MPa)	Force (N)	Hinge strain
10 mm compression	−11.636%	40.070°	85.38%	−0.04270	94.74	11.96%
10 mm tension	+11.636%	14.806°	53.49%	+0.08238	182.76	18.04%
Full closure	−18.350%	45.000°	100.00%	—	—	17.81%
Fully open	+15.470%	0.000°	50.00%	—	—	35.63%

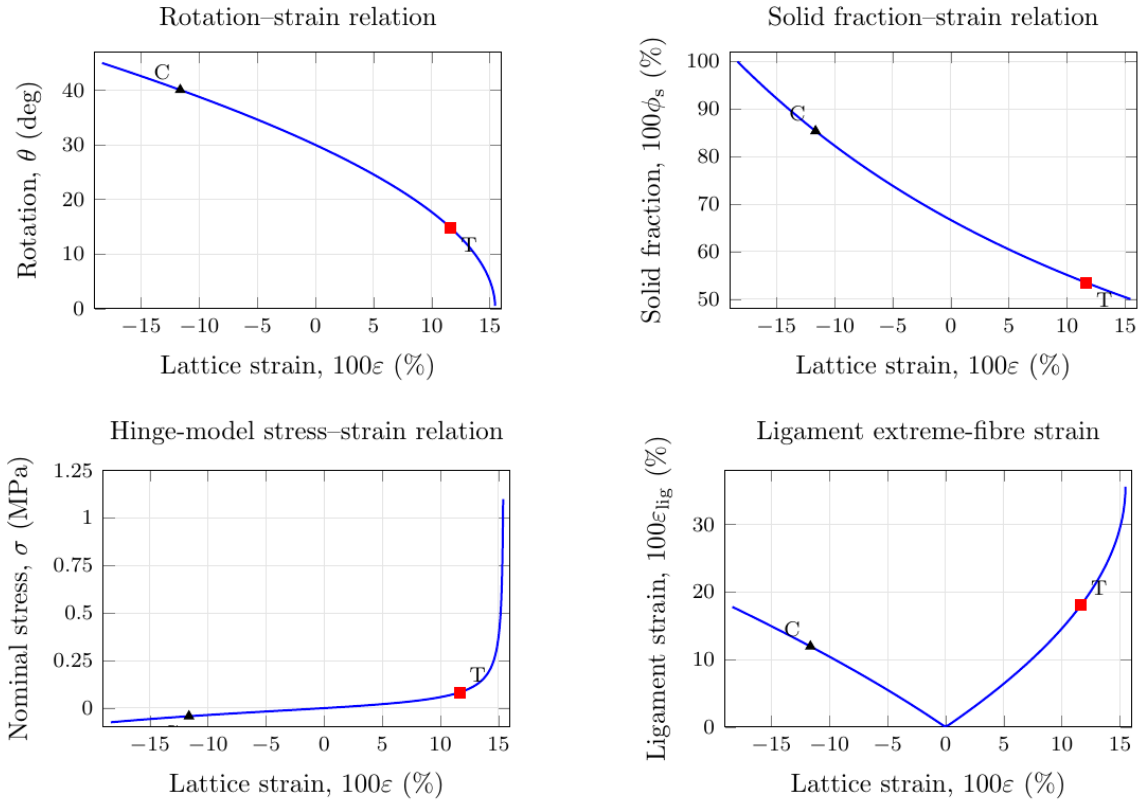


Figure 6. Parametric analytical curves for rotation, solid area fraction, hinge-model stress and ligament strain. C and T identify the 10 mm compression and tension cases.

At full closure, edge contact governs further deformation. At the fully open limit, Eq. (5) is singular. Stress and force estimates are therefore omitted at both endpoints. The solid fraction here is a two-dimensional area fraction.

6 ANSYS simulation method

Table 4. Static Structural loading and output settings.

Item	Completed compression case
Solver	ANSYS Mechanical 2025 R1, Static Structural
Loading	Lower end fixed. Upper end prescribed 5.00 mm toward −Y.
Reaction probe	Global Y component at the fixed support
Result set	Load step 1, substep 3, end time 1.0 s

The prescribed setup specified an isotropic linear-elastic TPU model with provisional E = 26 MPa and material ν = 0.45. It called for free transverse motion at the loaded end and large deflection enabled. The ANSYS exports verify the support scope and the solved fields. The saved project is needed to confirm the material assignment, transverse constraints, large-deflection setting and mesh statistics.

6.1 Gauge locations and strain convention

Probes 1 and 2 are the lower and upper faces of the interior axial gauge. Probes 3 and 4 are the left and right faces of the interior transverse gauge. Both initial centre-to-centre spacings are 11.09 mm. Each probe reports the signed maximum of its selected displacement component over its face at the end of the load step.



Figure 7. Interior gauge positions and initial spacings. Schematic with spacing annotations.

$$\Delta H = u_{Y,2} - u_{Y,1}, \quad \Delta W = u_{X,4} - u_{X,3}. \quad (7)$$

$$\nu_{eff} = -(\Delta W / W_{g,0}) / (\Delta H / H_{g,0}). \quad (8)$$

The two gauge strains describe the interior lattice. The prescribed end travel applies to the full specimen, including the end regions.

7 ANSYS results

7.1 Final substep and probe calculation

Table 5. Directional displacements at the final result set.

Probe	Location and component	Signed displacement
1	Lower interior gauge, Y	−1.0075 mm
2	Upper interior gauge, Y	−2.0737 mm
3	Left interior gauge, X	+0.56920 mm
4	Right interior gauge, X	−0.17464 mm

$$\Delta H = (-2.0737) - (-1.0075) = -1.0662 \text{ mm} \quad \Delta W = (-0.17464) - (+0.56920) = -0.74384 \text{ mm}$$

$$\varepsilon_Y = -1.0662/11.09 = -9.614\%, \quad \varepsilon_X = -0.74384/11.09 = -6.707\%, \quad \mathbf{\nu_{eff} = -0.698}.$$

Table 6. Summary of the completed 5 mm simulation.

Simulation output at end time	Value
Axial reaction at fixed support (+Y)	29.895 N
Specimen secant stiffness, reaction / end travel	5.979 N/mm
Maximum total deformation	5.8474 mm
Global X deformation range	−2.1369 to +2.0743 mm
Global Y deformation range	−5.7920 to +0.17963 mm
Peak von Mises stress	2.9744 MPa

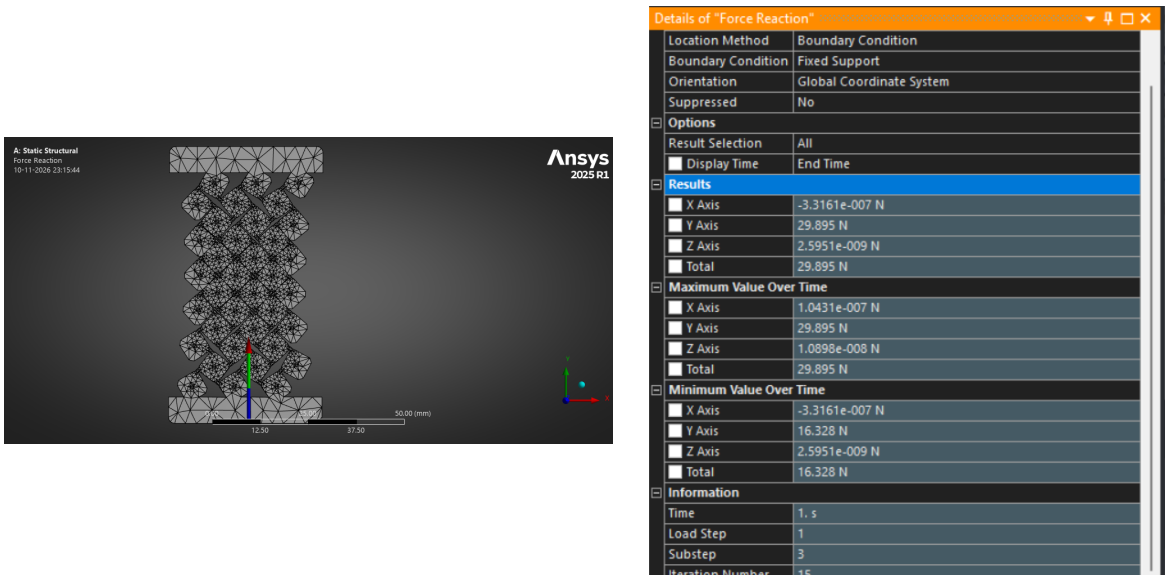


Figure 8. Fixed-support reaction in the ANSYS model (left) and the final Y reaction of +29.895 N (right).

7.2 Directional displacement fields

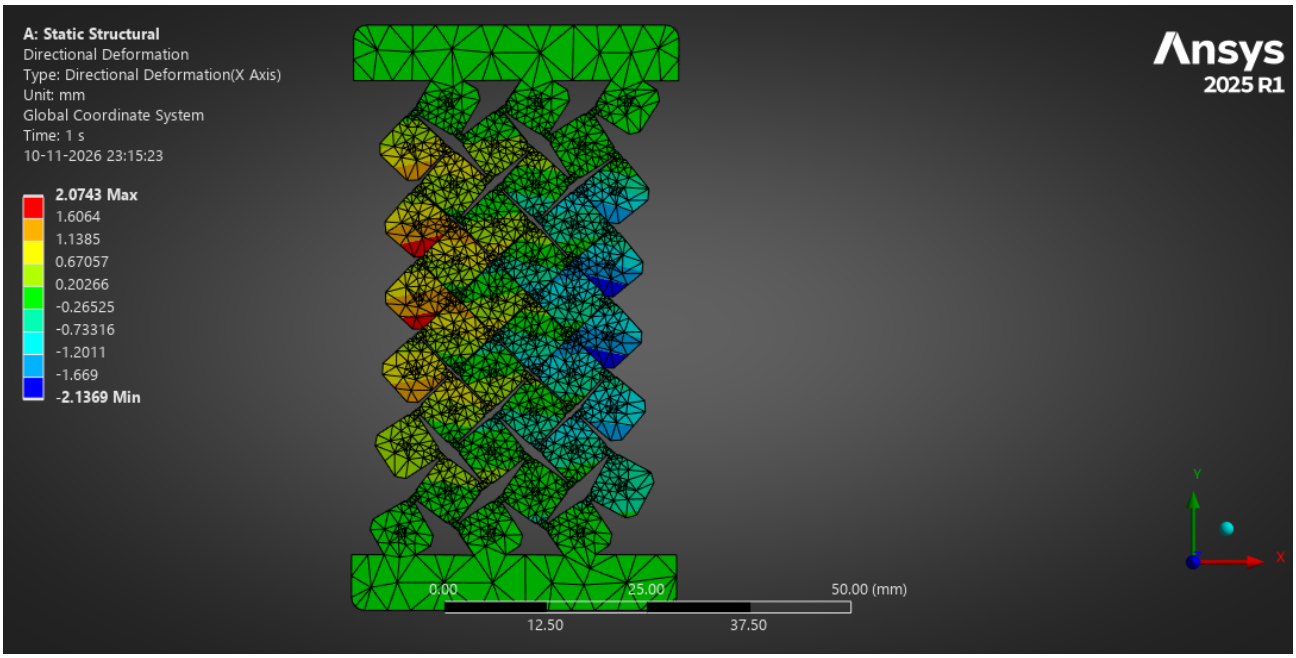


Figure 9. X displacement at end time. The central left side moves toward +X, and the central right side moves toward −X. These directions show inward motion.

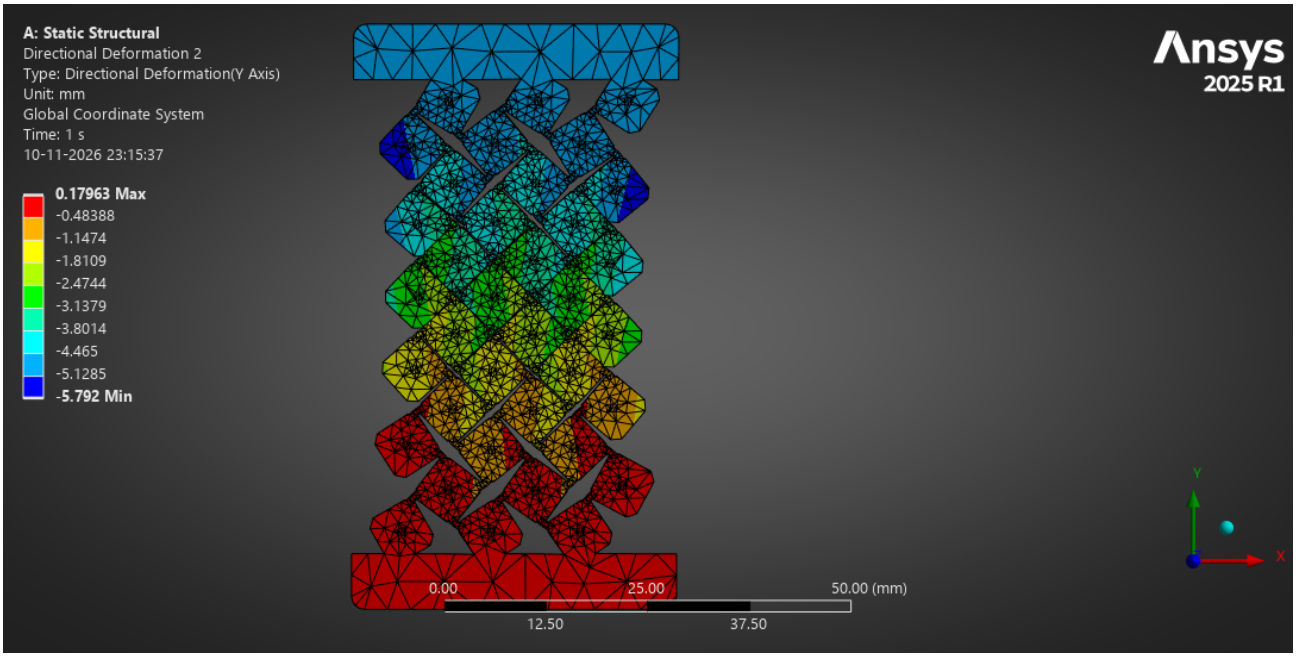


Figure 10. Y displacement at end time. The loading direction is −Y. The plot reports local displacements throughout the specimen.

7.3 Total deformation and stress

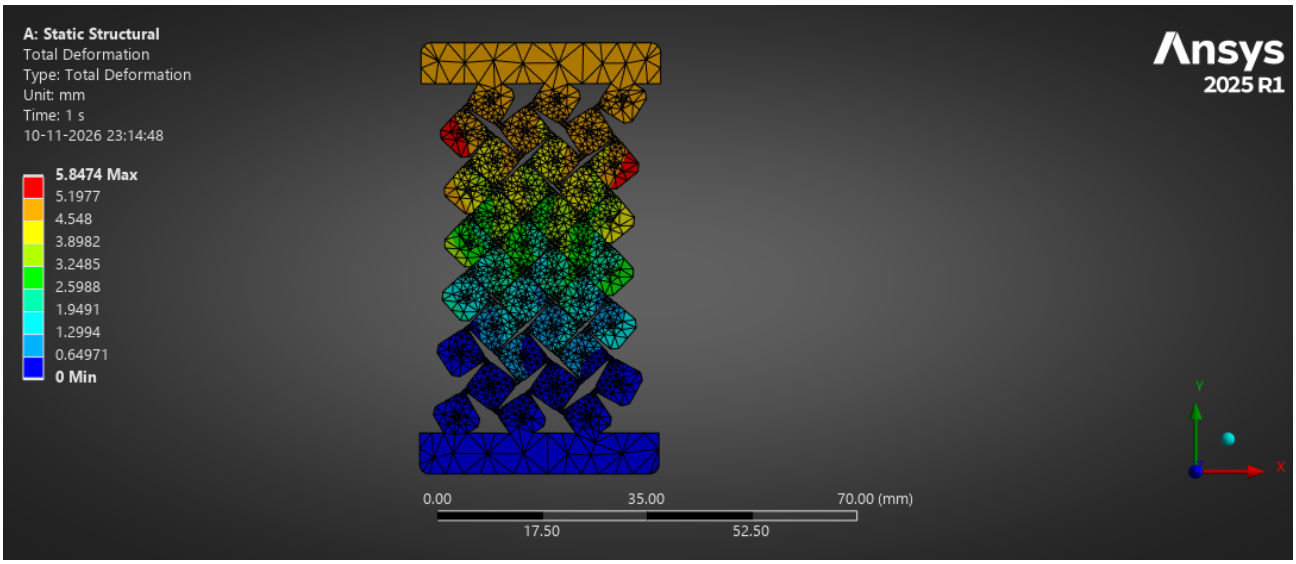


Figure 11. Total deformation at end time. The maximum of 5.8474 mm is the magnitude of the local displacement vector. The prescribed axial end travel is 5.00 mm.

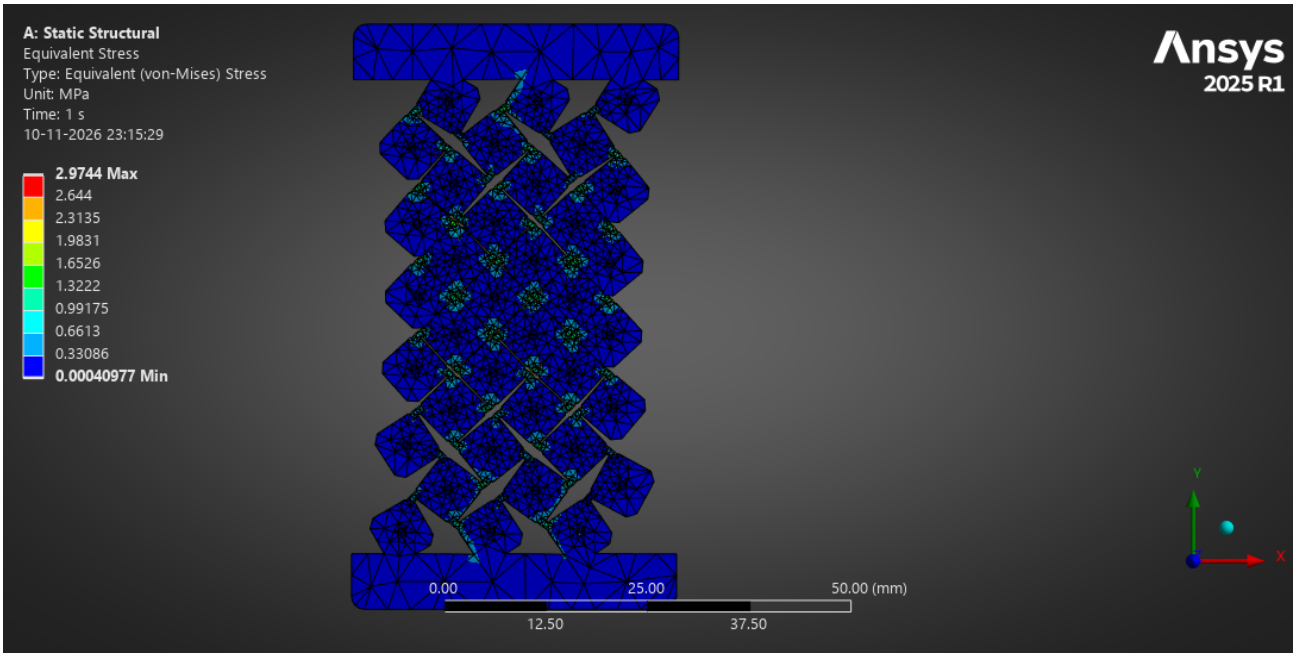
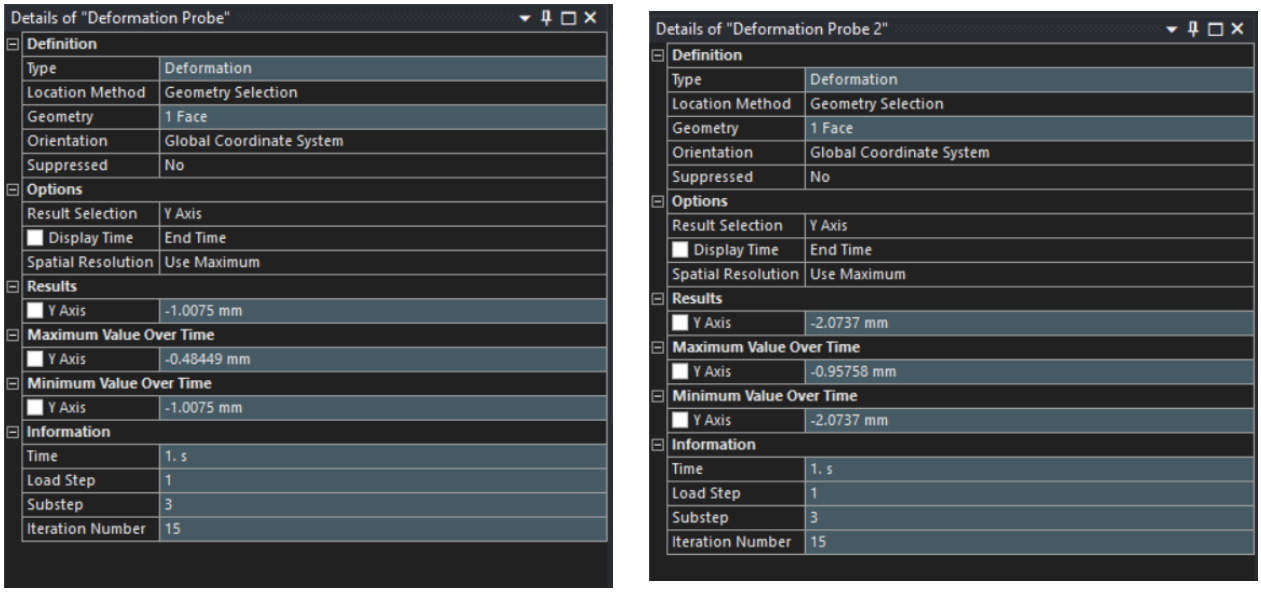


Figure 12. Equivalent von Mises stress at end time. The displayed peak is 2.9744 MPa.

7.4 Deformation probe outputs



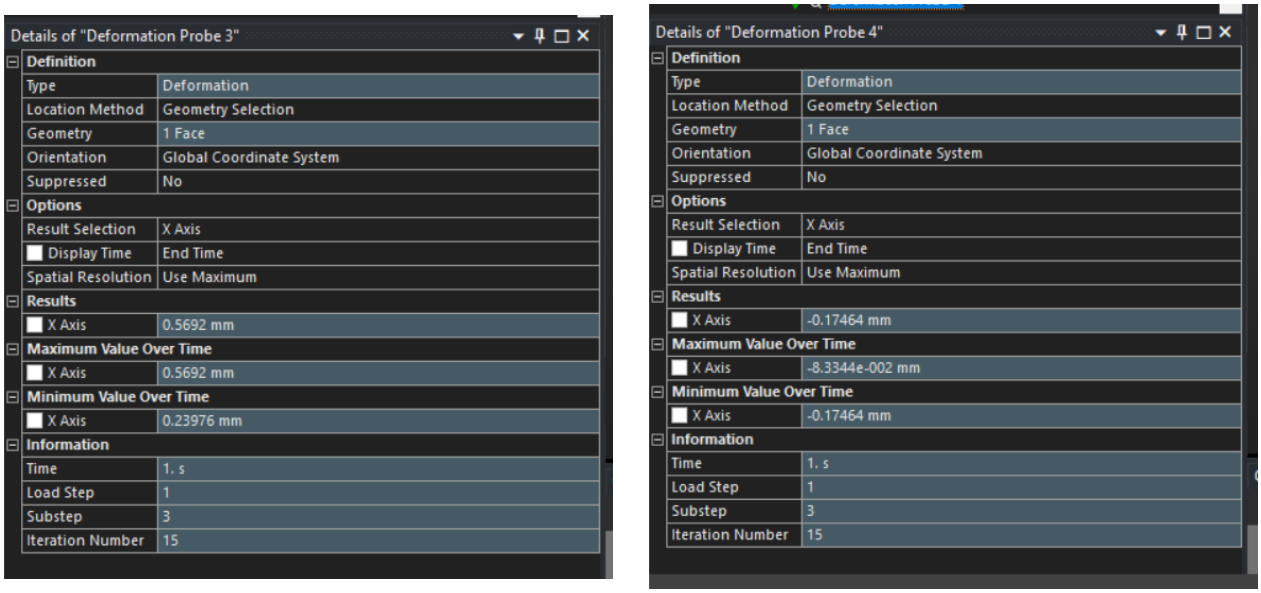
Details of "Deformation Probe"

Definition	
Type	Deformation
Location Method	Geometry Selection
Geometry	1 Face
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	Y Axis
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Y Axis	-1.0075 mm
Maximum Value Over Time	
Y Axis	-0.48449 mm
Minimum Value Over Time	
Y Axis	-1.0075 mm
Information	
Time	1. s
Load Step	1
Substep	3
Iteration Number	15

Details of "Deformation Probe 2"

Definition	
Type	Deformation
Location Method	Geometry Selection
Geometry	1 Face
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	Y Axis
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
Y Axis	-2.0737 mm
Maximum Value Over Time	
Y Axis	-0.95758 mm
Minimum Value Over Time	
Y Axis	-2.0737 mm
Information	
Time	1. s
Load Step	1
Substep	3
Iteration Number	15

Figure 13. End-time axial probe outputs for the lower and upper gauge faces.



Details of "Deformation Probe 3"

Definition	
Type	Deformation
Location Method	Geometry Selection
Geometry	1 Face
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	X Axis
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
X Axis	0.5692 mm
Maximum Value Over Time	
X Axis	0.5692 mm
Minimum Value Over Time	
X Axis	0.23976 mm
Information	
Time	1. s
Load Step	1
Substep	3
Iteration Number	15

Details of "Deformation Probe 4"

Definition	
Type	Deformation
Location Method	Geometry Selection
Geometry	1 Face
Orientation	Global Coordinate System
Suppressed	No
Options	
Result Selection	X Axis
Display Time	End Time
Spatial Resolution	Use Maximum
Results	
X Axis	-0.17464 mm
Maximum Value Over Time	
X Axis	-8.3344e-002 mm
Minimum Value Over Time	
X Axis	-0.17464 mm
Information	
Time	1. s
Load Step	1
Substep	3
Iteration Number	15

Figure 14. End-time transverse probe outputs for the left and right gauge faces.

8 Discussion and conclusions

8.1 Comparison with the analytical model

The rigid-unit model predicts equal axial and transverse strains and v = −1. In the simulated specimen, the transverse strain magnitude is about 70% of the axial strain magnitude. Finite square and ligament compliance, the number of cells and end-block restraint can contribute to this difference [2, 6]. The individual contributions were outside the scope of this simulation.

The analytical forces use the nominal 50 mm extrusion depth, a 10 mm active-region displacement and an active reference area. The ANSYS reaction uses 5 mm end travel. A numerical force comparison requires matching geometry, loading and reference area. For a future 5 mm-deep slice, multiplying its force by ten assumes uniform response through the nominal 50 mm depth. The corresponding gross stress convention uses 50 × 50 mm², while the analytical hinge model uses 44.37 × 50 mm².

8.2 Limitations

The probe-based ratio is a local secant estimate. Face rotation can shift the location of the reported maximum away from the face centre. The stress peak is mesh-sensitive near the ligaments. Material calibration, mesh convergence, large-deformation contact, rate dependence, cyclic response and physical validation remain unverified.

8.3 Related applications

Related work examines hierarchical rotating units, numerical tailoring of rotating-square and rotating-rectangle patterns, and the use of auxetic lattices in tubular structures [6–9].

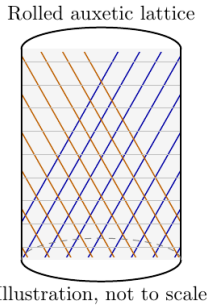


Figure 15. Conceptual rolled lattice. This illustration is outside the simulated scope.

8.4 Conclusion

At 5.00 mm of prescribed compression, the simulated lattice shortened and narrowed. The support reaction was **29.895 N**, and the interior gauges gave an effective Poisson's ratio of **−0.698**. The sign of the strain ratio supports the predicted auxetic response for this load case.

9 Analytical calculation check

The original MATLAB script reproduces the nominal analytical load cases in Section 5. It uses millimetres, newtons, MPa and radians internally.

```
% Analytical arithmetic check for the rotating-squares model
clear; clc;

% Planned geometry and provisional TPU modulus
 a = 8.0; % square side, mm
 d = 1.2; % chamfer distance, mm
 h = 1.0; % ligament width, mm
 t = 50.0; % specimen depth, mm
 E = 26.0; % provisional TPU modulus, MPa [VERIFY]
 theta0 = pi/6;

% Hinge properties
Lh = sqrt(2)*d*cos(theta0);
I = t*h^3/12;
kh = E*I/Lh;
Eeff = E*h^3/(3*Lh*a^2*sin(theta0)^2);

% Reference active lattice dimensions and force area
H0 = 85.94; % mm
W0 = 44.37; % mm
A0 = W0*t; % mm^2

eps10 = 10/H0;
labels = {'10 mm compression','10 mm tension'};
epsList = [-eps10, +eps10];

fprintf('Lh = %.5f mm, kh = %.3f N mm/rad\n', Lh, kh);
fprintf('Eeff = %.4f MPa\n\n', Eeff);
for n = 1:numel(epsList)
    eps = epsList(n);
    theta = acos((1 + eps)*cos(theta0));
    solidFraction = 1/(2*cos(theta)^2);
    deltaTheta = theta - theta0;

    sigma = -(cos(theta0)/sin(theta)) * ...
        (4*kh*deltaTheta/(a^2*t*cos(theta)^2)) * ...
        (1 + deltaTheta*tan(theta));

    force = abs(sigma)*A0;
    ligamentStrain = abs(deltaTheta)*h/Lh;

    fprintf('%s\n', labels{n});
    fprintf(' strain = %+8.5f, theta = %7.3f deg\n', ...
        eps, theta*180/pi);
    fprintf(' solid fraction = %6.2f %%\n', 100*solidFraction);
    fprintf(' stress = %+8.5f MPa, force = %7.2f N\n', ...
        sigma, force);
    fprintf(' ligament surface strain = %6.2f %%\n\n', ...
        100*ligamentStrain);
end

% Ideal kinematic result
nuIdeal = -1;
fprintf('Ideal in-plane Poisson ratio = %.1f\n', nuIdeal);
```

The geometric limit rows follow by setting θ to 0° or 45° in Eqs. (2) and (6). The ANSYS reaction and probe displacements are read directly from the result exports.

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Project records. Rotating-square theory report and STEP geometry audit, November 2026. ANSYS Mechanical 2025 R1 force, deformation and stress exports dated 10 November 2026.